Fundamentals of Foundation Models
Summer 2026
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Tutorial Set 1
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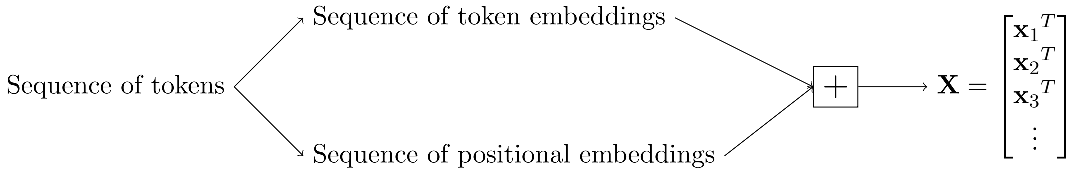
1 Recap GPT

Design parameters of GPT

- `dim_embd` (denote as d)
- `vocab_size` (denote as v)
- `context_length` (denote as c)
- `n_heads`
- `n_layers`

1.1 Input embeddings



Q1: Dimensions of $\mathbf{X}$? $\mathbf{X} \in \mathbb{R}^{c \times d}$, where each token $x_i^T \in \mathbb{R}^d$ is a column vector of dimension $d \times 1$.
Q2: Num. of learnable parameters in the input embedding? $dv + dc$

1.2 Transformer block

$$\mathbf{X} \longrightarrow \boxed{\text{Transformer block}} \longrightarrow \tilde{\mathbf{X}}$$

Q3: Dimensions of transformer output $\tilde{\mathbf{X}}$? Same as input $\tilde{\mathbf{X}} \in \mathbb{R}^{c \times d}$

1.3 Output embedding

Q4: Dimensions of output of output embedding? $\mathbb{R}^{c \times v}$ (each token is mapped to a probability distribution over all entries in the vocabulary)
Q5: Num. of learnable parameters? Normally, the number of parameters is dv, **but** with weight tying (sharing), we reuse the input embedding matrix, so no additional parameters are needed.

1.4 Attention layer

Query, key, value matrices $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{c \times d}$ are computed from input $\mathbf{X} \in \mathbb{R}^{c \times d}$ as

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \qquad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \qquad \mathbf{V} = \mathbf{X}\mathbf{W}_V, \tag{1}$$

with learnable weight matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d}$. Queries and keys are combined as $\mathbf{Q}\mathbf{K}^T \in \mathbb{R}^{c \times c}$ and the final output of attention (without masking) is computed as

$$\operatorname{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}. \tag{2}$$

2 Problems

Problem 1 (Softmax scaling)**.** In general, softmax(x) has very small gradients if x is large, which makes it difficult to optimize. In this problem we motivate the scaling with $1/\sqrt{d}$ used in the attention mechanism in Equation (2). Assume all entries in $\mathbf{Q}, \mathbf{K}$ are i.i.d. with $\mathcal{N}(0, 1)$.

1. Compute $\mathbb{E}\left[\mathbf{Q}\mathbf{K}^T\right]$?
2. Compute $\operatorname{Var}[\mathbf{Q}\mathbf{K}^T]$ and $\operatorname{Var}[1/\sqrt{d}\mathbf{Q}\mathbf{K}^T]$?

Problem 2 (Shift equivariant of transformers)**.** Consider the input

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1{}^T \\ \mathbf{x}_2{}^T \\ \mathbf{x}_3{}^T \end{bmatrix},$$

consisting of three tokens and its shifted version

$$S(\mathbf{X}) = \begin{bmatrix} \mathbf{x}_2{}^T \\ \mathbf{x}_1{}^T \\ \mathbf{x}_3{}^T \end{bmatrix},$$

with shift operator $S()$ that switches the position of the first and second token. We denote applying an attention layer (without masking) to the input $\mathbf{X}$ as

$$A(\mathbf{X}) = \operatorname{softmax}\left(\mathbf{X}\mathbf{W}_Q\mathbf{W}_K{}^T\mathbf{X}^T\right)\mathbf{X}\mathbf{W}_V,$$

where we dropped the scaling factor to shorten notation.

1. Show that
$$S(A(\mathbf{X})) = A(S(\mathbf{X})).$$
2. What happens if we use masked attention as discussed in class?